

PeriSense Health Risk Prediction API

Base URL: <https://perisense-ml-model.onrender.com>

The API provides maternal health risk prediction using a **Random Forest machine learning model**. It accepts basic maternal health measurements and returns the predicted risk level, confidence score, and probability distribution across the three risk categories.

POST /predict

Predicts the maternal health risk level for a patient.

Request Body:

```
{
  "Age": 1,
  "SystolicBP": 1,
  "DiastolicBP": 1,
  "BS": 1,
  "BodyTemp": 1,
  "HeartRate": 1
}
```

Field	Description
Age	Patient's age
SystolicBP	Systolic blood pressure
DiastolicBP	Diastolic blood pressure
BS	Blood Sugar
BodyTemp	Body Temperature
HeartRate	Heart Rate

Diastolic Diastolic blood pressure

BP

BS Blood sugar level

BodyTemp Body temperature

HeartRate Heart rate

Response — 200 OK:

```
{
  "risk_level": "low risk",
  "confidence": 0.5781,
  "probabilities": {
    "high risk": 0.01,
    "low risk": 0.5781,
    "mid risk": 0.4119
  }
}
```

Response Fields

- **risk_level** — The predicted maternal health risk category: **low risk**, **mid risk**, or **high risk**.
- **confidence** — The model's confidence in the predicted risk category, represented as a decimal between 0 and 1.
- **probabilities** — The model's probability distribution across all three risk categories.

Error Response

The API returns **422 Unprocessable Entity** when the submitted request fails input validation.

API Documentation

Interactive Swagger/OpenAPI documentation is available at:

[PeriSense API Documentation](#)

OpenAPI specification: [openapi.json](#)